

Energy-Flow Cosmology, Part 2: Field Equations and Observable Mapping (μ , Σ , S_8)

Morten Magnusson

Symbiose Research, Sandnes, Norway

ORCID: 0009-0002-4860-5095

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Abstract. This is the second paper in the EFC whitepaper series. Part 1 established that EFC contains Λ CDM as a limiting case in the perturbation sector under three independent recovery conditions. Here we derive the field equations from first principles, define the entropy field $S(a)$ and its coupling to the effective gravitational parameters $\mu(k, z)$ and $\Sigma(k, z)$, and establish the complete observable mapping to the growth rate $f\sigma_8$, weak-lensing amplitude S_8 , and the regime response surface $R(k, S)$. All equations are referred to the sealed blind predictions (NUTS cycles, 2026-02-18 and 2026-02-21) and to the validated observational results from DES Y6, KiDS-1000, DESI DR2, and BOSS DR12.

Companion paper: Magnusson (2026a), “EFC: Recovery Conditions and the Λ CDM Limit,” doi:10.6084/m9.figshare.31135597.

Introduction

Part 1 of this series established the structural foundation of EFC: the existence of three analytically independent recovery conditions under which EFC reduces to Λ CDM/GR, and the containment relation $\text{EFC} \supset_{\text{pert.}} \Lambda\text{CDM}$.

The present paper descends from that structure to the level of concrete physics: *how does the entropy field couple to gravity, and what are the observable consequences?*

We proceed in three steps. First, we derive the EFC field equations from the entropy-gradient principle. Second, we define the full observable mapping $(\mu, \Sigma) \rightarrow (f\sigma_8, S_8)$. Third, we connect each equation to its validation status in the EFC Validation Ledger.

The Entropy Field and Coupling Structure

Fundamental principle

EFC postulates that the effective gravitational response is not constant but is modulated by the local entropy structure of the universe. The central scalar is the entropy field:

Definition 1 (Entropy field $S(a)$). $S(a)$ is a dimensionless scalar field tracking the mean entropy content of structure at cosmic epoch a . It satisfies $S \in [0, 1]$, with $S \rightarrow 0$ in the early universe (L0/L1 regime, GR limit) and $S \rightarrow S_{\max} \lesssim 1$ in the late-time structure-dominated era (L2 regime).

The field $S(a)$ encodes the degree to which entropy-gradient corrections are active. In the L0/L1 regime ($S < S_c \approx 0.1$), all EFC corrections vanish identically (Recovery Theorem 2, Part 1).

Derived coupling formula

The core equation linking $S(a)$ to the effective gravitational coupling is:

$$\mu(a) = 1 + \beta S(a), \quad (1)$$

with $\beta \approx 0.16$ from SPARC calibration. Equation (1) is the *derived* coupling formula: β is not a free parameter but is constrained by the structural properties of the model.

In the sigmoid-gate parameterisation used for CMB-compatible fits:

$$\mu(a) = 1 - B g(a; n), \quad g(a; n) = \frac{1}{1 + (a_t/a)^n}, \quad (2)$$

where $B > 0$ enforces $\mu < 1$ (weakened growth), a_t is the transition epoch, and n sets the sharpness. Equations (1) and (2) are consistent: the sigmoid form is the epoch-parameterised version of the entropy coupling once $S(a) \propto -g(a; n)$ in the growth era.

A Gaussian parameterisation used for MCMC sampling:

$$\mu(a) = 1 + \beta \exp\left[-\frac{(a - a_t)^2}{2\sigma_a^2}\right]. \quad (3)$$

The EFC kernel

At the scale-dependent level, the full EFC kernel is:

$$\mu_{\text{EFC}}(k, z) = [1 - \bar{S}(z)] + \frac{\tilde{s}(k, z)}{\tilde{\delta}(k, z)}, \quad (4)$$

where $\bar{S}(z)$ is the spatially averaged entropy field, and $\tilde{s}(k, z)/\tilde{\delta}(k, z)$ is the Fourier-space entropy-to-density ratio encoding scale dependence. In the limit $\bar{S} \rightarrow 0$ and $\tilde{s} \rightarrow 0$, equation (4) reduces to $\mu_{\text{EFC}} = 1$, recovering GR (cf. Recovery Theorem 2).

Field Equations

Entropy-gradient form

The EFC field equation at the metric level takes the form:

$$G_{\mu\nu} = f(\nabla S, E_f), \quad (5)$$

where ∇S is the entropy gradient and E_f is the local energy-flow density. In the L0/L1 limit ($\nabla S \rightarrow 0$), equation (5) reduces to the standard Einstein tensor $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$.

Modified growth equation

The growth rate $f \equiv d \ln D / d \ln a$ satisfies:

$$f' + f^2 + \left(\frac{1}{2} - \frac{3}{2} \tilde{\Omega}_m \right) f = \frac{3}{2} \mu(a) \tilde{\Omega}_m, \quad (6)$$

where primes denote $d/d \ln a$, $\tilde{\Omega}_m = \Omega_m(a)/E^2(a)$, and $\mu(a)$ enters from equation (2). When $\mu = 1$, equation (6) reduces to the standard Λ CDM growth ODE.

The $\mu < 1$ condition (weakened gravity in the growth era) suppresses structure formation relative to Λ CDM, generating the observed S_8 tension reduction.

Background parameterisation

The background expansion is parameterised as:

$$E^2(z) \rightarrow E^2(z) [1 + \alpha_{L2} f(z) \Theta(z)], \quad (7)$$

with the regime-transition function:

$$\Theta(z) = \frac{1}{2} \left[1 + \tanh \left(\frac{z_{L1L2} - z}{\Delta z} \right) \right]. \quad (8)$$

The background α_{L2} -gate was found to be inconsistent with joint CMB+BAO constraints and is accordingly absent from the current phenomenological model. EFC operates in the perturbation sector only (see Part 1, Remark 1).

Observable Mapping

Growth rate $f\sigma_8$

The predicted growth rate observable is:

$$[f\sigma_8](z) = f(z) \sigma_8(z) = f(z) \sigma_{8,0} D(z)/D(0), \quad (9)$$

where $D(z)$ is the linear growth factor obtained by integrating equation (6) with $\mu(a)$ from equation (2).

Empirical result: Sealed prediction (NUTS, 2026-02-18): $f\sigma_8(z = 0.7) = 0.430$ vs. Λ CDM prediction 0.449 (2.0 σ deviation). LOO robustness: 7/7 folds pass; $|\alpha|/\sigma \in [1.84, 2.42]$. Full signal: $\alpha_{L2} = -1.00 \pm 0.46$ (2.20 σ , $\Delta\text{AIC} = -2.91$).

Lensing amplitude S_8

The S_8 suppression integral is:

$$\Delta\sigma_8 \propto (1 - \mu_0) \int g(a; n) d\ln a, \quad (10)$$

yielding:

$$S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3} \approx S_{8,\Lambda\text{CDM}} - \delta S_8(\mu_0, B, n). \quad (11)$$

Empirical result: Pre-registered prediction P3 (DES Y6, lensing-to-CMB S_8 ratio): PASS at 0.3 σ (T2 tier, data-likelihood level).

Regime response surface $R(k, S)$

The full observable space is spanned by the regime response surface:

$$R(k, S) : (k, S) \mapsto G_{\text{eff}}(k, S)/G_N, \quad (12)$$

mapping wavenumber k and structural parameter S to the effective gravitational coupling ratio. Each dataset (DES, KiDS, DESI, BOSS, SPARC) provides an empirical slice through $R(k, S)$ at a specific (k, z) coordinate.

Interpretation: Equation (12) unifies the perturbation-sector observables into a single geometric object. Multi-probe consistency requires that all empirical slices lie on the same surface, which is the content of the cross-dataset validation programme.

Weak lensing: $\Sigma(k, z)$

The lensing slip parameter is:

$$\Sigma(k, z) = \frac{\mu_{\text{lens}}(k, z)}{G_N}, \quad (13)$$

where μ_{lens} is the lensing-weighted effective coupling. The structural constraint “pure $\mu < 1$ excluded” (T3, Validation Ledger) requires $\Sigma \neq \mu$ in the L2 regime, i.e. a gravitational slip $\eta \equiv \Psi/\Phi \neq 1$ is present. This is directly testable with Stage-IV lensing surveys.

Validation Table

Table 1 maps each equation to its observational status.

Table 1: Observable mapping: equation–data correspondence

Observable	Equation	Dataset	Result	Status
α_{L2} (background)	(7)	CMB+BAO	COLLAPSED	Failed
α_{L2} (growth)	(9)	DESI+BOSS	-1.00 ± 0.46 (2.20σ)	Pass
$f\sigma_8(z=0.7)$	(9)	RSD	0.430 vs 0.449 (2.0σ)	Sealed
$f\sigma_8$ LOO	(9)	RSD	7/7 folds	Pass
S_8 (DES Y6)	(11)	DES	0.3σ (pre-reg.)	Pass
BAO transfer	(7)	DESI→BOSS	No refit needed	Pass
Lensing (KiDS)	(13)	KiDS-1000	Regime-activated fit improved	Pass
$\mu < 1$ sign	(2)	Multi-probe	$B > 0$ required	Pass
Pure $\mu < 1$ excl.	(13)	Structure	Slip required ($\Sigma \neq \mu$)	T3
Solar system (PPN)	(1)	KT1	GR recovered ($< 10^{-5}$)	Pass
$w(a) = -\beta(S) \cdot a$	(13)	Euclid	Candidate: $w_0 \approx -1$	Planned

Table 2: Open observational tests (planned)

Test	Observable	Survey
Scale-dependent $\mu(k, z)$	k^* EFC characteristic scale	Euclid, Rubin LSST
$\sigma_8(z)$ full trajectory	Growth history	Stage-IV
ISW sign-flip in deep voids	$\mathcal{C}$ metric	DESI+CMB cross
Gravitational slip η	Σ/μ	Euclid
Cosmic dipole anisotropy	Regime-dependent β	Future all-sky

Discussion

The degeneracy problem and its resolution

Parameter α_{L2} (and B, μ_0) cannot be uniquely determined from fewer than three simultaneous independent probes (identifiability gate IG-1). This is a structural consequence of the μ – Σ degeneracy valley in the Planck+BAO posterior.

The resolution is multi-probe locking: when DES (S_8), DESI/BOSS ($f\sigma_8$), and KiDS (Σ) are combined simultaneously, the degeneracy is broken. The current 2.20σ signal on α_{L2} is therefore a *lower bound* on significance; Stage-IV surveys will either confirm or falsify it.

Structural hierarchy

The equations in this paper form a strict hierarchy:

1. Entropy field $S(a)$ [fundamental]
2. Coupling formula $\mu(a) = 1 + \beta S(a)$ [derived]
3. Growth equation with $\mu(a)$ [dynamical]
4. $f\sigma_8, S_8, \Sigma$ observables [empirical]
5. $R(k, S)$ response surface [unifying]

Each level is derived from the one above. This hierarchy connects the ontological foundation (entropy-gradient coupling) to the observable predictions without free parameters beyond $\beta \approx 0.16$ and the transition parameters $(a_t, n, \Delta z)$.

What is not derived from first principles

The following remain phenomenological inputs, not derived:

- The functional form of $S(a)$ (assumed smooth, monotone in $L2$)
- The transition redshift z_{L1L2}
- The full relativistic action (EFT ansatz established, full derivation pending; see [figshare.31562200](https://figshare.com/figure/31562200))

These are stated explicitly as open problems.

Conclusions

We have derived the complete chain from the EFC entropy field $S(a)$ to the observables $f\sigma_8$, S_8 , Σ , and $R(k, S)$. The core equation $\mu(a) = 1 + \beta S(a)$ connects the fundamental entropy-gradient principle to the modified growth equation, which in turn generates testable predictions on all current and next-generation cosmological surveys.

The observable mapping is internally consistent with the three recovery conditions established in Part 1: setting $S \rightarrow 0$ (or equivalently $\beta \rightarrow 0$) returns all observables to their Λ CDM values without residual corrections.

The 2.20σ signal in the growth sector, robust across 7/7 LOO folds and consistent across DES, KiDS, DESI, and BOSS, constitutes the current empirical core of EFC. It is not a detection. It is a persistent directional signal that the falsifiability protocol ([figshare.31251277](https://figshare.com/figure/31251277)) will resolve with Stage-IV data.

Series structure. This is Part 2 of the EFC whitepaper series. Part 1 (Recovery & Limits) established the theoretical boundary conditions. Part 3 (Data, Validation, and Falsification Protocol) will present the full structured validation ledger and kill criteria.

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